

Usual Stability Loan

A Mechanism for Leveraging USD0++

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June 15, 2025

Introduction

Following the introduction of the price floor, the Usual ecosystem is launching a new feature, *Usual Stability Loans* (USL), which allows users to leverage USD0++ using its underlying collateral to capitalize on drops in USD0++ prices. Users will be able to borrow USD0 against the collateral of USD0++ at a fixed rate—independent of secondary market pricing—when USD0++ deviates from parity. This borrowed USD0 can then be iteratively used to purchase USD0++, enabling users to gain leveraged exposure to USD0++ and earn compounded USUAL rewards.

It is important to note that the borrowed USD0 will be collateralized by eUSD0 (Euler USD0) held in the collateral treasury wallet.

These loans will be offered at a fixed interest rate, creating a more competitive market compared to the USD0++ market on Morpho. Ultimately, the Usual Stability Loan introduces a new dimension to the ecosystem by unlocking the USD0 behind USD0++, thereby expanding opportunities for exposure and rewards within the USD0++ ecosystem.

Concept

The Usual Stability Loan (USL) will operate entirely through an Euler vault, allowing users to borrow USD0 using USD0++ as collateral. This mechanism enables users to leverage USD0++ by borrowing the underlying collateral, USD0, at a fixed rate and recursively looping USD0++ to amplify their USUAL rewards.

The vault is structured so that only the DAO will provide liquidity, allocating USD0, which will be represented by an associated eUSD0 (Euler USD0) position corresponding to vault share token from the deposit in the vault. This eUSD0 will become a new accepted collateral by the protocol. The vault will have a fixed interest rate, Loan-to-Liquidation Value (LLTV), and Maximum Loan-to-Value (LTV), all of which will be set by the DAO. Additionally, the exchange rate used for USD0++ to USD0 will be hardcoded as 1, meaning users will be able to borrow the same value regardless of secondary market pricing. Importantly, the vault will be ungoverned, meaning that Usual will manage it, and all parameters except the oracle will be immutable once set.

Borrowers will access leverage based on a DAO-determined LLTV, combined with a fixed, DAO-set interest rate. This structure results in a fixed maturity period before potential liquidation, which currently stands at 3.4 years—the remaining lock duration of USD0++ at the time of writing. This setup ensures that:

- Borrowers can access leverage at fixed rates that may be more favorable (i.e., lower) than those offered by alternative lending protocols such as Morpho and Aave.
- The protocol (i.e., the DAO Treasury) secures a fixed margin through interest payments by the end of

maturity, guaranteeing revenue for the DAO.

- No undercollateralization occurs in a zero-volatility context, where USD0 remains fully backed 1:1 by USYC or equivalent assets.

Furthermore, it is important to note that the DAO can only allocate USD0 to the vault from the treasury. This is because only USD0 that is not being used to collateralize USD0++ can be allocated. This restriction is enforced by the smart contract infrastructure, which strictly prevents any scenario in which USD0++ could become undercollateralized.

Vault Setup

The vault parameters are defined as follows:

- **Max LTV:** Set to 0.83, meaning a user locking 1 USD0++ can borrow up to 0.83 USD0, as USD0++ has a fixed conversion rate of 1:1 to USD0. If the collateral, net of accrued interest, reaches this threshold, the user will be liquidated. In such a case, all USD0++ will be burned, and any remaining USD0 will be allocated to the DAO Treasury.
- **LLTV (Liquidation Loan-to-Value):** Set to 0.9999, meaning that when a user's borrowed amount, including accrued interest, reaches this threshold, the DAO is authorized to liquidate the position.
- **Interest Rate:** Fixed at 5%. A user locking 1 USD0++ will accrue interest debt at a rate of 0.05 USD0 per year. Consequently, if a user borrows 0.83 USD0 at inception, the total debt (principal plus interest) will grow to exactly 1.0 USD0 after four years, making the locked USD0++ fully liquidatable. Additionally, this rate is set to be lower than those in Morpho markets, ensuring the vault remains competitive and profitable.
- **Market Type:** Defined as ungoverned. Usual will not engage with curators for this vault, as all parameters are immutable and require no further risk management.

Flow

The mechanism can be understood from the DAO and user perspective (where the user flow can be both a generalized manner or by using hypothetical examples).

0.1 DAO Flow

The following flow outlines the deposit process and eventual withdrawal of the position:

1. **Locking:** The DAO mint new USD0 to deposit into the vault. It receives eUSD0 on a 1:1 basis for each USD0 deposited, eUSD0 is used as collateral by the protocol to mint new USD0.
2. **Maintenance:** The eUSD0 remains in the treasury for the duration of the loan.
3. **Exiting:** Upon exit, depending on the type of exit, the DAO either collects the accrued interest from repayment or liquidates the user, burning the respective eUSD0 in the process.
 - (a) **User Repayment:** If the user repays the loan, the accrued interest paid to unlock the collateral is allocated to the treasury. The initially lent USD0 is returned in full, and the corresponding eUSD0 is burned.
 - (b) **Liquidation:** If the user fails to repay and the loan is liquidated, the liquidation profit (denoted in USD0)—while the corresponding eUSD0 is burned—is allocated to the DAO. Additionally, the USD0++ collateral is allocated to the DAO, which has the option to burn it.

0.2 Example, of Generalized User Flow

Values for illustration: $LTV = 0.83$, $LLTV = 0.9999$, Interest rate = 5%. The following generalized flow shows the different outcomes for a user using the Usual Stability Loan:

1. **Locking:** User obtains 1 USD0++ either locking 1 USD0 for 1 USD0++ or via swapping on the secondary market at a discounted price.
2. **Borrowing:** User borrows 0.83 USD0 per USD0++.
3. **Interest Accrual:** User accrues an interest debt of 0.05 USD0 per USD0++ collateralized per year.
4. **Exit:** User can either let the loan sit and be liquidated or can pay the accrued interest plus the original borrowed USD0.
 - (a) **Liquidation:** Following liquidation, the DAO burns the USD0++ corresponding to the non-returned and the remaining USD0++ along with its respective USD0 are allocated to the DAO treasury.
 - (b) **Repayment:** Following repayment of the loan, the user receives the USD0++ collateral, any USD0 interest paid is allocated to the DAO treasury and the USD0 repaid for the original loan is reallocated to the DAO Collateral Smart Contract.

Example User Flows

Values for illustration: $LTV = 0.83$, $LLTV = 0.9999$, Interest rate = 5% APR for simplicity. The following three examples cover the different scenarios that can occur after a user borrows USD0 with USD0++:

Example 1: Borrower is Liquidated at Maturity

1. **Locking:** Alice locks 1 USD0 into USD0++.
2. **Borrowing:** She borrows 0.83 USD0 (at 83% LTV).
3. **Interest Accrual:** Over 4 years, the 0.83 USD0 debt accumulates interest until it equals 1 USD0.
4. **Liquidation:** At maturity, Alice fails to repay. The protocol seizes her locked USD0++ (worth 1 USD0) and:
 - **Burns 0.83 USD0++** (the original principal minted).
 - **Captures 0.17 USD0** (from USD0++) as interest (sent to the DAO treasury).
 - **Returns 0 USD0** to Alice (she is fully liquidated).
5. **Result:** The protocol has zero bad debt (fully covered), and 0.17 USD0 of revenue is realized by the DAO.

Example 2: Borrower Repays Before Maturity

1. **Locking and Borrowing:** Same as above; Alice locks 1 USD0 into USD0++, borrows 0.83 USD0.
2. **Partial Interest Accrual:** Suppose after 2 years, her total owed debt is 0.93 USD0 (principal plus partial interest).
3. **Repayment:** Alice decides to repay the 0.93 USD0:

- She buys or otherwise acquires 0.93 USD0 from the market.
 - She pays 0.93 USD0 back to the protocol, which burns 0.83 as principal and captures 0.10 as accrued interest so far.
 - The protocol unlocks her 1 USD0 from USD0++ and returns it to her.
4. **Result:** Alice saves some portion of the not-yet-accrued interest, and the protocol still makes a profit of 0.1 USD0.

Example 3: Borrower Loops and Repays Before Maturity

1. **Locking and Borrowing:** Alice locks 1 USD0 into USD0++, borrows 0.83 USD0.
2. **Iteration:** Using the 0.83 USD0, Alice locks this USD0 for 0.83 USD0++ and then uses this to borrow 0.6889 USD0.
3. **Partial Interest Accrual:** Suppose after 1 year, her total owed debt is 1.594 USD0 (principal plus partial interest). The first loan accrued a debt of 0.88 USD0 while the second loan accrued a debt of 0.714 USD0.
4. **Repayment:** Alice decides to repay 0.714 USD0:
 - She buys or otherwise acquires 0.714 USD0 from the market and uses it to unlock her 0.83 USD0++.
 - She then swaps the received USD0++ to USD0 and mints additional USD0 to repay the remaining 0.88 USD0 owed in interest. This allows her to unlock the 1 USD0++ first used as collateral to borrow.
 - The protocol captures 0.07221 USD0 as interest.
5. **Result:** Alice saves some portion of the not-yet-accrued interest, and the protocol still makes a profit of 0.07221 USD0.

Example 4: Bob Borrows and Redeems (No Further Looping)

Below is an illustration of Bob's actions and the protocol's final accounting. We compare:

- **Scenario A (No Borrow):** Bob simply locks 1 USD0 in USD0++ for 4 years, and the protocol earns a 4% annual yield on 1 USYC.
- **Scenario B (Borrow and Redeem):** Bob locks 1 USD0 in USD0++, borrows 0.83 USD0 at 5% rate, and immediately redeems the borrowed USD0 for 0.83 USYC. The protocol is left with 0.17 USYC (plus Bob's locked 1 USD0++).

Key Rates and Assumptions:

- **Collateral Yield on USYC:** 4% annually (simple interest).
- **PSM Borrow Rate:** 5% annually (simple interest).
- **Time Horizon:** 4 years.
- **All yield on USYC** goes to the protocol's treasury.

Based on this, we can analyze the protocol's revenue when users either hold USD0++ until maturity or borrow and redeem.

Parameter	Scenario A: No Borrow	Scenario B: Borrow & Redeem
Initial USYC locked	1.00	1.00
Bob borrows \$	0.00	0.83
Bob redeems borrowed \$	N/A	0.83 USDC or USYC
Vault USYC left	1.00	0.17
USYC yield after 4 yrs (4%/yr)	1.00 → 1.16	0.17 → 0.20
Bob's final debt at 5%/yr	0.00	≈ 1.00 USD0
Protocol interest from Bob	0.00	≈ 0.17 USD0
Protocol net yield (USYC side)	+0.16	+0.03
Protocol net interest (stable side)	0.00	+0.17
Total protocol gain	≈ 0.16 USYC	≈ 0.17 (<i>stable</i>) + 0.03 (<i>USYC</i>)

Table 1: Comparison of Protocol Outcomes: 4-Year Horizon

Accountability and No Undercollateralization:

- 1. Immediate Redemption of Borrowed USD0:** When Bob redeems 0.83 USD0, the protocol *burns* those 0.83 USD0 and hands Bob 0.83 USYC from the vault, leaving 0.17 USYC behind.
- 2. Locked USD0 Cannot Unlock Without Repayment:** Bob's 1 USD0++ is *inaccessible* unless he repays the 0.83 USD0 (plus interest).
- 3. End State (Repay or Default):**
 - **If Bob Repays:** He brings back 1.00 USD0, which the protocol burns for the principal + interest. The vault can then rebalance and let Bob unlock his 1 USD0 (or its equivalent).
 - **If Bob Defaults:** The protocol seizes the locked 1 USD0++ and burns the redeemed portion. Since the borrowed portion (0.83) was already burned at redemption, no extra stable remains in circulation, so there is no shortfall.
- 4. Conclusion:** In normal condition, the protocol is never left with more USD0 claims than USYC collateral.

Thus, even when Bob “exits” by redeeming the borrowed amount, the *locked USD0++* and forced repayment (or liquidation) mechanism ensure the system remains fully collateralized. Meanwhile, the DAO earns a higher effective return (interest + partial yield) than in the simple no-borrow scenario.

Risks and Limitations

The following risks are inherited with the launch of this model:

- **Volatility of Underlying Collateral (USYC):** If USYC is not perfectly stable, high LTVs can lead to undercollateralization. The system relies on USYC holding a 1:1 peg.
- **Zero-Cushion at Liquidation:** Setting LTV close to 1 (e.g., 0.83 going to 1) leaves no surplus for borrowers if they fail to repay, which might reduce willingness to borrow.
- **Liquidity Crunch Risks:** If large holders redeem USD0 suddenly, the system must have enough USDC on hand. In a perfect 1:1 scenario, this is fine, as the amount of USD0 should be aligned with the amount of USDC directly redeemable.
- **Third Party Risk:** Risk of Hack on the third parties.
- **USD0++ price:** Potential Risk of impermanent undercollateralization if the price goes below the LTV, however this is handled by the DAO.

Conclusion and Future Extensions

In conclusion, the USL mechanism introduces a new dimension of leverage within the USUAL ecosystem, creating opportunities for both borrowers and the DAO. Borrowers benefit from fixed-rate leverage, often at lower rates than external lending protocols like Morpho, while maintaining a straightforward and predictable interest structure. At the same time, the DAO captures a fixed margin from interest, ensuring a sustainable source of revenue. The system remains safeguarded against undercollateralization, provided that **USYC** holds its peg and the protocol enforces strict repay and burn requirements. This structure not only enhances capital efficiency but also strengthens the overall ecosystem by enabling new avenues for profitability and expansion.